

9. Edge Linking

Edges detected: [1, 0, 1, 0, 1]

(a) Interpretation of the discontinuity problem

The sequence has gaps (0s) between detected edge pixels (1s), meaning the edge detector failed to mark every point along the true boundary — likely due to noise or weak gradient response at those positions. A broken edge cannot reliably define a closed object boundary.

(b) Justification for the linking method

Edge linking connects nearby edge pixels that share similar gradient direction and magnitude (or uses morphological closing / Hough-based grouping) to bridge small gaps and reconstruct continuous boundaries from fragmented edge detections.

(c) Apply linking step

Gap pixels (0) that lie between two edge pixels (1) and are consistent with the edge direction are filled in (set to 1) to restore continuity.

(d) Computed result

Position	1	2	3	4	5
Before linking	1	0	1	0	1
After linking	1	1	1	1	1

(e) Interpretation of continuity

After linking, the edge sequence becomes fully continuous [1, 1, 1, 1, 1], representing a complete, unbroken boundary suitable for further shape analysis or object outlining.